



# WAN Authentication — Assignment

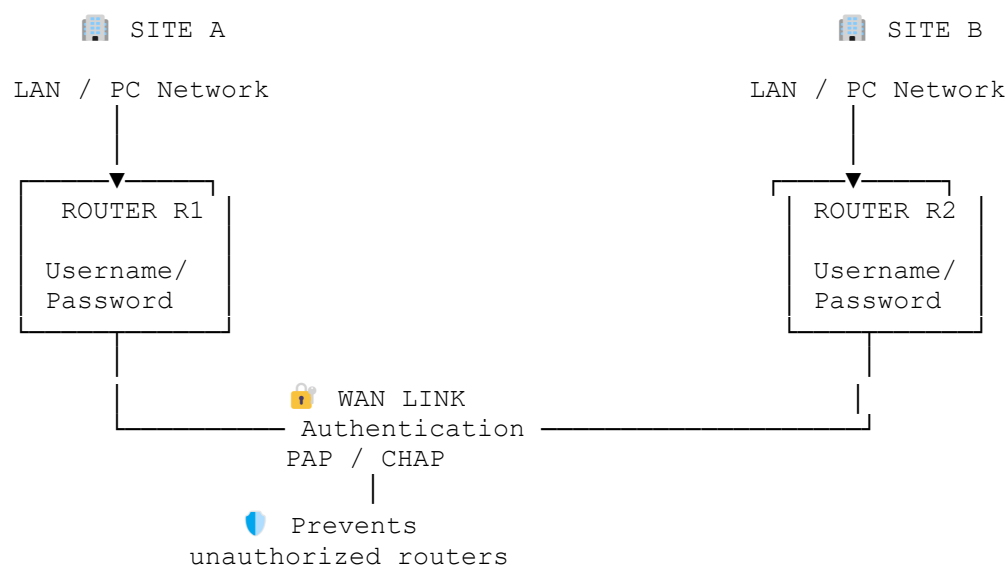
## Question 1

Draw a simple network diagram showing two routers connected over a WAN link and label where authentication is needed to prevent unauthorized access.

### Answer

A WAN connection allows two geographically separated routers to communicate.

**Authentication is required at the WAN connection between the two routers** so that only an authorized router can establish the connection.



### 📌 Where is authentication needed?

Authentication is needed **between R1 and R2 on the WAN link**. Protocols such as **PAP** and **CHAP** can verify the identity of the remote router before allowing the WAN connection.

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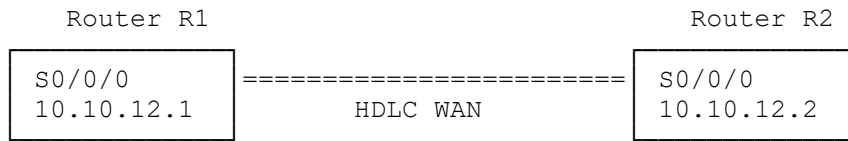
## Question 2

Configure a sample HDLC WAN connection between two routers using Packet Tracer or any network simulator, and explain why HDLC does not provide authentication.

### Answer

**HDLC (High-Level Data Link Control)** is a Layer 2 WAN protocol used to encapsulate data over a serial connection. Cisco routers use Cisco HDLC as the default encapsulation on many serial interfaces.

## Simple Topology



## R1 Configuration

```
R1> enable
R1# configure terminal
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.10.12.1 255.255.255.252
R1(config-if)# encapsulation hdlc
R1(config-if)# no shutdown
R1(config-if)# end
```

## R2 Configuration

```
R2> enable
R2# configure terminal
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.10.12.2 255.255.255.252
R2(config-if)# encapsulation hdlc
R2(config-if)# no shutdown
R2(config-if)# end
```

## If R1 is the DCE side

```
R1(config)# interface serial 0/0/0
R1(config-if)# clock rate 64000
```

## Verification

```
R1# show interfaces serial 0/0/0
```

You should see that the interface and line protocol are **up**.

You can also test connectivity:

```
R1# ping 10.10.12.2
```

## Why doesn't HDLC provide authentication?

**HDLC provides data-link framing and encapsulation, but it does not provide built-in user/device authentication such as PAP or CHAP.** Therefore, simply configuring HDLC does not verify the identity of the remote router.

### Security point:

There is no username/password authentication command in the HDLC configuration:

```
encapsulation hdlc
```

Unlike PPP, HDLC does not have commands such as:

```
ppp authentication pap
```

or

```
ppp authentication chap
```

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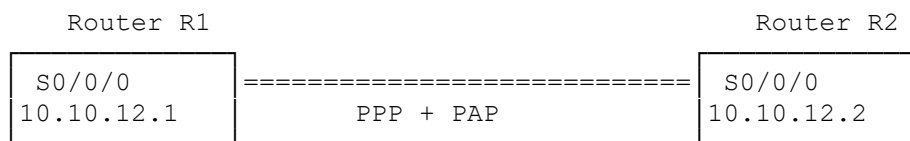
## Question 3

**Set up a PPP connection between two routers in Packet Tracer and enable PAP (Password Authentication Protocol) authentication. Show the configuration commands and explain what happens if the password is incorrect.**

### Answer

**PPP (Point-to-Point Protocol)** supports authentication and is commonly used on point-to-point WAN links. **PAP** authenticates a router using a username and password.

#### Topology



#### R1 Configuration

```
R1> enable
R1# configure terminal

R1(config)# username R2 password cisco123

R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.10.12.1 255.255.255.252
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication pap
R1(config-if)# ppp pap sent-username R1 password cisco123
R1(config-if)# no shutdown

R1(config-if)# end
```

#### R2 Configuration

```
R2> enable
R2# configure terminal

R2(config)# username R1 password cisco123
```

```
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.10.12.2 255.255.255.252
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication pap
R2(config-if)# ppp pap sent-username R2 password cisco123
R2(config-if)# no shutdown

R2(config-if)# end
```

### Verify the Connection

```
R1# show interfaces serial 0/0/0
```

Then test:

```
R1# ping 10.10.12.2
```

If the usernames and passwords match correctly, the PPP connection should establish successfully.

### What happens if the password is incorrect?

If the password configured on one router does not match the password expected by the other router, **PAP authentication fails**. The PPP link will not successfully establish its authenticated connection, and communication across the WAN link will fail.

### Important PAP Security Point

PAP uses a **two-way username/password exchange**, making it less secure than CHAP because the password is sent without the challenge-response protection used by CHAP.

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## Question 4

**Modify your PPP configuration to use CHAP (Challenge Handshake Authentication Protocol) instead of PAP, and describe one key difference in how CHAP improves security over PAP.**

### Answer

**CHAP** provides authentication using a **challenge-response mechanism**. Instead of simply sending the password as part of the authentication exchange, the authenticating router sends a challenge and the other router responds using a calculated value based on the shared secret.

### Change PAP to CHAP

Remove PAP configuration if necessary:

```
R1(config)# interface serial 0/0/0
R1(config-if)# no ppp authentication pap
```

Configure CHAP:

### R1

```
R1(config)# username R2 password cisco123

R1(config)# interface serial 0/0/0
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication chap
```

### R2

```
R2(config)# username R1 password cisco123

R2(config)# interface serial 0/0/0
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication chap
```

### Verify

```
R1# show interfaces serial 0/0/0
```

Then:

```
R1# ping 10.10.12.2
```

### Key Difference: PAP vs CHAP

| Feature                  | PAP                                                                                    | CHAP                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Authentication method    | Username/password exchange                                                             | Challenge-response                                                                      |
| Password protection      | Less secure                                                                            | More secure                                                                             |
| Challenge mechanism      |  No |  Yes |
| Password repeatedly sent | Yes                                                                                    | No                                                                                      |
| Security                 | Lower                                                                                  | Higher                                                                                  |

### How CHAP Improves Security

**PAP sends authentication credentials in a simple exchange, while CHAP uses a challenge-response process so the actual password is not sent across the link as part of the authentication exchange.** This makes CHAP a stronger authentication method for PPP WAN connections.

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## Question 5

Research and list two real-world scenarios (such as public WiFi login or mobile data authentication) where WAN authentication protocols like PAP, CHAP, or MS-CHAP are used, and explain why authentication is critical in each case.

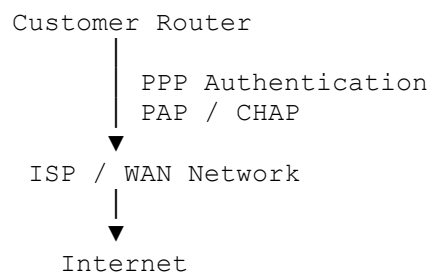
## Answer

### Scenario 1: ISP / Broadband Internet Access

Internet service providers can use **PPP-based authentication**, historically including PAP or CHAP, to authenticate subscribers connecting to the provider's network.

#### Why authentication is important

Authentication allows the provider to determine whether the subscriber is authorized to use the service. It can also associate the connection with the correct customer account and apply appropriate service policies.

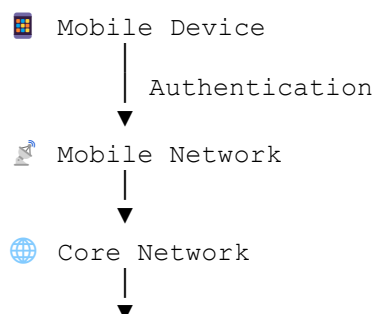


### Scenario 2: Mobile Network Authentication

Mobile networks use authentication mechanisms to verify subscribers and devices before providing access to network services. Modern cellular systems generally use **stronger cellular-specific authentication mechanisms**, rather than relying directly on classic PAP/CHAP.

#### Why authentication is important

Authentication prevents unauthorized devices or subscribers from accessing network services and helps the mobile operator identify legitimate subscribers. It is an essential part of protecting the operator's network and customer services.





## Quick Revision

|  Protocol | Main Purpose                         | Authentication                                                                                                        |
|--------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>HDLC</b>                                                                                | WAN data encapsulation               |  No built-in PAP/CHAP authentication |
| <b>PPP</b>                                                                                 | Point-to-point WAN communication     |  Supports authentication             |
| <b>PAP</b>                                                                                 | Username/password authentication     |  Yes                                 |
| <b>CHAP</b>                                                                                | Challenge-response authentication    |  Yes                                 |
| <b>MS-CHAP</b>                                                                             | Microsoft PPP authentication variant |  Yes                                 |

### Final Takeaway

**HDLC provides WAN encapsulation but does not provide built-in authentication. PPP supports authentication through protocols such as PAP and CHAP, with CHAP providing a stronger challenge-response mechanism than PAP.**